When does data quality help human deferral? Knowledge-based systems for AI governance and managerial decision-making

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Short communication.

Abstract

AI governance must allocate scarce review capacity when data-quality rules prohibit automation. We investigate when quality information improves deferral beyond model probabilities, separating estimation, admission and rule enforcement. A fixed-loss allocator handles mandatory overload exactly; propensity-weighted fitting addresses selective review logs under explicit overlap assumptions. Following exploratory credit experiments, we froze a validation plan before analyzing a new Bank Marketing dataset. In its primary contrast, adding quality only to the automation-loss head reduced cost by 5.23% relative to probability-only estimation under informative missingness (20/20 splits; corrected 95% interval for absolute reduction, 0.0099–0.0224; $p < 0.001$). This supports a conditional benefit, not universal superiority. Across six planned secondary comparisons, inverse-propensity weighting did not significantly improve cost after Holm adjustment. A separate transfer check uses recorded human image classifications without simulating responses to corrupted stimuli. Policy ablations distinguish visible-defect coverage from zero rule violations; service simulations distinguish review priority from capacity under unit service times. The contribution is a reproducible conditions analysis of governance-constrained deferral, with matched component comparisons and an independently evaluated quality hypothesis.

Keywords: Knowledge-based systems; Learning to defer; Data quality; AI governance; Human–AI collaboration; Review capacity; Managerial decision-making

1 Introduction

Organizations divide decisions between models and reviewers. Governance assigns responsibilities and controls [1, 2, 3, 4], but oversight alone does not establish effective control [5]. Deteriorating data can increase mandatory review; managers must distinguish predictive benefit, rule coverage and service capacity.

Rejection and learning to defer trade model error against abstention or reviewer error [6, 7, 8, 9]; uncertainty methods offer simpler alternatives [10, 11]. Triage separates model and human error [12]. DeCCaF already combines costs, limited feedback and workload constraints [13]; related work addresses sparse expert predictions [14], quality-aware cascades [15] and congestion [16]. We test quality inputs under rules that prohibit automation and require holds under overload, claiming neither a new surrogate nor the first constrained deferral system.

Completeness and validity are measurable but incomplete proxies for quality [17, 18, 19]. Human-in-the-loop and knowledge-based decision systems connect predictions to action [20, 21, 22]; routing traces document decisions rather than explain the underlying model [23, 24, 25]. We ask: *under what data-quality and feedback conditions does learned deferral improve on probability-only estimation and cost-aware heuristics under mandatory review rules?* Contributions are a precise allocation model, matched component comparisons under selective feedback, and a planned quality validation on a previously unanalyzed business dataset. Negative results and rule-coverage failures are retained.

2 Method

2.1 Rules and scarce review capacity

A predictor gives p_i and automates with $\hat{y}_i = \mathbf{1}[p_i \geq 1/6]$. False-negative, false-positive, review and hold costs are 5, 1, $c_R = 0.05$ and $c_H = 1.5$. Observable quality $q_i = (u_i, v_i)$ is the missing-field and out-of-range fraction. Training-only bounds are $[Q_{.01,j} - .1t_j, Q_{.99,j} + .1t_j]$, $t_j = \max(Q_{.99,j} - Q_{.01,j}, 1)$. The knowledge base marks $m_i = \mathbf{1}[u_i \geq .5 \vee v_i \geq .25]$. These are explicit experimental policy assumptions, not regulatory thresholds or empirically optimal rules.

Let a_i, r_i, h_i denote automate, review and hold, $M = \{i : m_i = 1\}$, and $B = \lfloor bn \rfloor$. For fixed estimated automation and reviewer losses, allocate by

$$\min \frac{1}{n} \sum_i \left[a_i \hat{\ell}_i^A + r_i(\hat{\ell}_i^H + c_R) + h_i c_H \right], \quad \sum_i h_i = (|M| - B)_+, \tag{1}$$

subject to $a_i + r_i + h_i = 1$, $\sum_i r_i \leq B$, $a_i \leq 1 - m_i$ and $h_i \leq m_i$, with binary actions. The last equality gives mandatory cases service priority; it is a normative assumption, not an economic optimum over all possible policies.

Proposition 1. If $|M| \leq B$, review all mandatory cases and up to $B - |M|$ optional cases with largest positive $\hat{g}_i = \hat{\ell}_i^A - \hat{\ell}_i^H - c_R$. If $|M| > B$, review the B mandatory cases with smallest $\hat{\ell}_i^H$ and hold the rest. *Proof.* Optional review changes cost by $-\hat{g}_i/n$. Under overload, only the sum of reviewed human losses varies; $(|M| - B)c_H$ is constant. Exchange arguments establish both cases. Allocation takes $O(n \log n)$. Exactness is conditional on estimated losses, not a guarantee of minimum realized cost.

An immediate consequence is a structural null: under mandatory overload, changing only the automation-loss head cannot alter the optimal route when the reviewer head is fixed. Quality can then help only through reviewer-loss estimates or policy coverage, not optional automation ranking.

Proposition 2. With fixed mandatory flags, identical arrivals, unit review times, no abandonment and work-

conserving mandatory service, queue ordering leaves mandatory backlog unchanged. *Proof.* The recurrence $Q_{t+1} = (Q_t + A_t^M - B_t)_+$ depends on counts, not identities. This does not extend automatically to heterogeneous service times, changing flags or reviewer fatigue.

2.2 Loss estimation and selective feedback

With $w(y) = 5y + (1-y)$, targets are $L^A = w(y)\mathbf{1}[\hat{y} \neq y]$ and $L^H = w(y)\mathbf{1}[\text{review} \neq y]$. All gain-fitting rows have outcome labels, but only logged rows have a reviewer decision. Two boosted regressors estimate the losses. Probability inputs are (p, s), $s = \min(5p, 1-p)$; quality inputs add $(u, v, pu, pv, \hat{y}u, \hat{y}v)$. The primary ablation changes only the automation head, retaining the same probability-only reviewer head, admission rule and allocator.

Let S indicate logging with known propensity $\pi(W)$ for observed covariates W. Unweighted fitting minimizes logged-row squared error; inverse-propensity weighting (IPW) uses weights $1/\pi(W)$, normalized to mean one among logged rows [26]. Under $S \perp L^H \mid W$ and $\pi(W) > 0$, $\mathbb{E}[SR_f/\pi(W)] = \mathbb{E}[R_f]$, $R_f = \{L^H - f(Z)\}^2$. This standard identity identifies the population fitting risk even if representation Z omits some logging covariates. It does not ensure finite-sample improvement or correct unobserved selection, missing outcome labels or test-time shift.

3 Experimental design

3.1 Exploration and prospective validation

The exploratory credit grid uses German, Australian, Default and HMEQ [27, 28, 29, 30, 31], simulated reviewers and nine quality conditions. Its published-surrogate adaptations are retained as exploratory evidence; they are not newly tuned state-of-the-art comparisons. No learned router beat the best heuristic when that grid was pooled, and its 18 quality-ablation tests did not survive Holm adjustment.

Before fetching new validation data, we committed the primary contrast, secondary family and complete design (plan `656f527`, implementation `da41f59`). Default supplies development evidence; Bank Marketing supplies a new business-domain validation [32] (45,211 rows). We remove post-call `duration`; categorical mappings use predictor-training rows only, with unknown or unseen categories treated as missing. Historical subscription labels are real, but review outcomes and costs remain simulated. This is cross-domain validation of the routing experiment, not a credit-deployment study.

Feature-identical rows stay together in partitions of 45% training, 15% calibration, 20% gain fitting and 20% testing (seeds 2–21). Histogram gradient boosting (at most 120 iterations, 15 leaves) is logistically calibrated on clean/missing/outlier/mixed copies; gain rows receive the same augmentation. Heads choose three or seven leaves

by grouped 80/20 inner gain validation (100 iterations, rate .06, L2=5), then refit. No augmented copies cross inner folds; test outcomes never select models.

Conditions are clean, informative missingness, stale values and severe defects. Informative missingness multiplies a uniform $[0,.35]$ rate by 1.8 for positives and .6 for negatives. Stale continuous values are multiplied by $[.55,.90]$ on half the rows. Severe shift combines 35–75% missingness and 8–25% outliers. Reviewers have constant error .10 or AI-correlated error, $\min(.45,.05+.25\mathbf{1}[\hat{y}\neq y]+.10[1-|2p_{clean}-1|])$. These mechanisms are illustrative; neither reviewer is defined from quality inputs.

3.2 Logging, matched comparisons and human transfer

Logging is random ($\pi=.4$), uncertainty-prioritized or rule-prioritized. Selective policies use $\pi=.1+.8\sigma(\alpha+z)$ with mean .4, where z is standardized threshold proximity or $4m+2(u+v)$. Shared uniforms and reviewer outcomes give methods identical logs. Comparators are probability-only, quality-in-automation-only, quality-in-both-heads, observed-feature two-head, single-head gain, confidence, Chow and rules-only. The feature comparator includes raw inputs and quality; every head has the same two-option search.

The matched arm shares probability-only reviewer-loss ordering under overload. A fill-budget arm removes optional admission cutoffs. Native, gain-order and no-rule arms isolate allocation and rules at 10% and 20% capacity.

CIFAR-10H supplies recorded human judgments [33]; ten-class choices are mapped to animal/vehicle. The predictor uses 8×8 averaged RGB features on 10,000 original images. Gain fitting samples one recorded choice per image; held-out empirical human error supplies evaluation loss. No corruption is applied: people saw clean images. Logging, head search and 20% capacity are unchanged. This is neither finance-expert validation nor a test of the primary quality hypothesis.

3.3 Endpoints and inference

The single primary test compares quality-in-automation-only with probability-only on Bank under informative missingness, random logging and 20% capacity, averaging reviewers per split. Success requires positive reduction, two-sided $p<.05$ and at least 1% relative improvement. Corrected resampled tests use ratio $.2/.8$ [34]; splits are dependent. Six secondary tests compare IPW versus unweighted full-quality fitting for two datasets and three logging policies, averaging conditions/reviewers at 20%, with one Holm adjustment [35]. Other analyses are exploratory; ambiguous earlier seven-/nine-test claims are retired.

4 Results and discussion

A conditional quality benefit was confirmed. The planned contrast reduced cost by 5.23% (absolute reduction 0.0161; corrected 95% interval [0.0099, 0.0224]; $p < .001$; 20/20 splits), meeting the 1% practical criterion (Table 2). Reviewer estimation and mandatory ordering are fixed; the fill-budget arm retains the same reduction. Figure 1 reports all condition/logging cells. Under mandatory overload, the automation-head effect is zero, as predicted. Confirmation concerns this synthetic mechanism on new data, not universal superiority.

Matched comparisons bound the claim. Table 1 uses a common matched policy. Quality in the automation head had the lowest mean among the matched variants on Default (0.5966) and Bank (0.4663); Chow cost risk gave 0.6018 and 0.4685, respectively. The original grid rules out universal superiority; lower means do not establish cross-setting significance.

Selective feedback did not make IPW uniformly useful. None of six secondary contrasts survives Holm adjustment (Table 2). Rule-prioritized logging selects approximately 89% of mandatory gain rows versus 40% under random logging at the same expected log budget. Weighting targets fitting risk, but variance and shift can offset benefits; changed losses need not change selected reviews. Human-transfer results are descriptive: on clean images under random logging, quality-in-automation-only cost was 0.1615, probability-only 0.1615, and Chow 0.1554; these means do not establish a human-domain quality effect.

Table 1: Expected cost under random logging and 20% capacity. Default/Bank average four conditions and two reviewers; CIFAR-10H uses clean images and empirical human error. Common mandatory ordering. Bold: lowest descriptive mean, not significance.

Method	Default	Bank	CIFAR-10H
Rules only	0.6703	0.5240	0.2778
Confidence	0.6039	0.4725	0.1670
Chow cost risk	0.6018	0.4685	**0.1554**
Single-head gain	0.6000	0.4682	0.1606
Probability-only two-head	0.6043	0.4706	0.1615
Quality in automation head	**0.5966**	**0.4663**	0.1615
Quality in both heads	0.5981	0.4674	0.1615
Observed-feature two-head	0.6030	0.4669	0.1579

Knowledge rules have costs and blind spots. Rules-only routing, no-rule learned routing and their combination separate policy enforcement from predictive benefit. Across Default and Bank, rules flagged 67.1% of records with

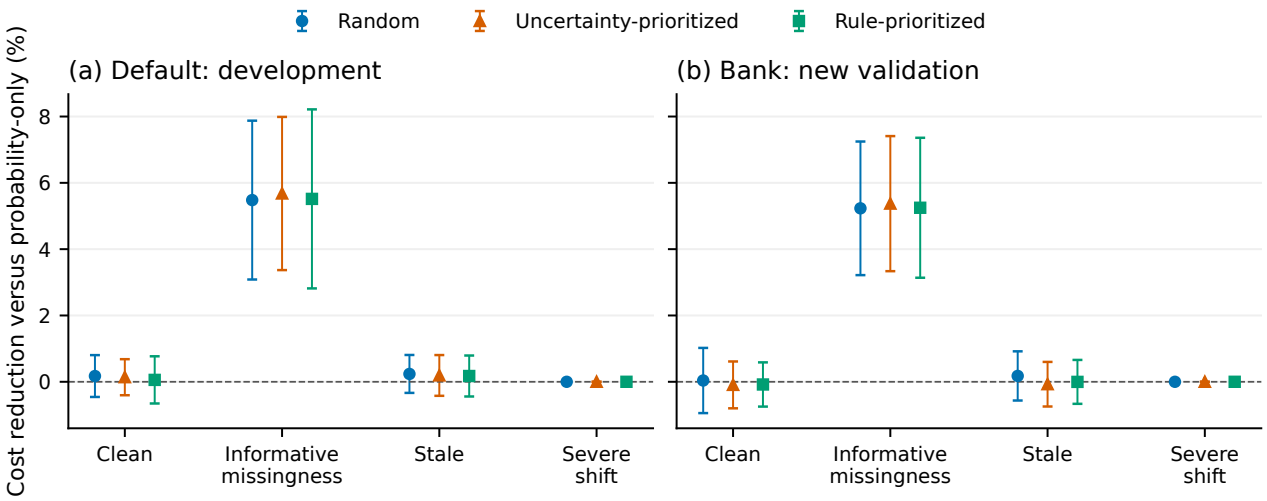


Figure 1: Quality in the automation head versus probability-only at 20% capacity, with common mandatory ordering. Positive reduction favors quality. Bars: approximate corrected 95% intervals over 20 splits, averaging reviewers. Only Bank/informative/random is confirmatory. Programmatically generated from saved outputs.

Table 2: Absolute cost reduction (positive favors the change), corrected 95% interval and p-value. The primary test stands alone; secondary p-values are Holm-adjusted across six contrasts.

Contrast	Reduction	95% interval	p
Primary: Bank, quality in automation head	+0.0161	[+0.0099, +0.0224]	< 0.001
IPW: Default, random	+0.0000	[+0.0000, +0.0000]	1.000
IPW: Default, uncertainty logs	+0.0007	[-0.0002, +0.0017]	0.641
IPW: Default, rule-prioritized logs	-0.0003	[-0.0018, +0.0012]	1.000
IPW: Bank, random	+0.0000	[+0.0000, +0.0000]	1.000
IPW: Bank, uncertainty logs	+0.0000	[-0.0008, +0.0008]	1.000
IPW: Bank, rule-prioritized logs	+0.0001	[-0.0009, +0.0011]	1.000

injected severe defects but 0.0% with injected stale values. Under severe shift, the combined router held 47.1% of all records; removing rules automated 51.9% of all records despite mandatory flags. A rule can be perfectly obeyed while missing stale but plausible values. These metrics concern the declared corruption and policy assumptions, not legal compliance or complete data-quality assurance.

Capacity results require their assumptions. The 40-day simulation introduces severe defects on days 11–20 and carries holds forward. Service counts all eligible mandatory arrivals, with unresolved cases unmet; quantiles account for censoring [36]. Cost ordering leaves 21.7% open at 20% capacity, none at 40%. At 40%, FIFO lowers the 95th-percentile delay from 13.9 to 7.1 days despite identical backlog counts (Fig. 2). This depends on unit

service and fixed flags.

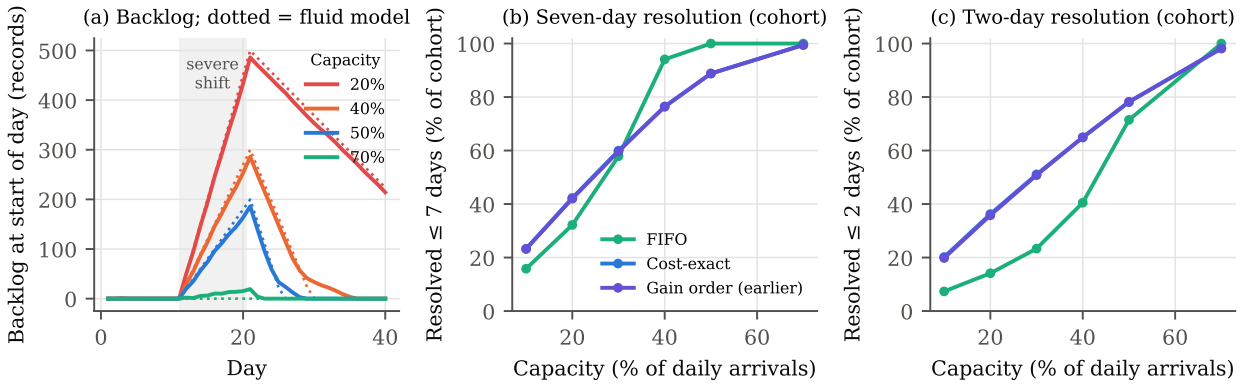


Figure 2: Credit simulation: (a) backlog, with dotted fluid approximation; (b,c) seven-/two-day resolution among eligible mandatory arrivals, unresolved cases unmet. Four datasets, two reviewers, ten seeds; unit service times. Programmatically generated from saved outputs.

Managers should validate quality features against a local cost-aware heuristic, document policy provenance, maintain exploration where review logs are selective, and monitor unresolved cohorts as well as rule violations. Organizational complementarity [37] and AI capability [38] motivate these decisions, but firm-performance or causal managerial effects are not tested here.

5 Limitations and conclusion

Bank is new to this project but old and synthetically corrupted. Human images do not validate finance workflows, damaged stimuli or fatigue. Costs and thresholds need domain validation; gradual drift [39], heterogeneous service, fairness and missing outcomes remain untested. IPW does not correct hidden selection. Published surrogate adaptations remain exploratory; no newly tuned DeCCaF comparison or new learning guarantee is claimed.

A frozen validation supports a conditional quality benefit, null weighting results and no universal superiority. Separating estimation, rules and capacity makes these boundaries observable.

Data and code availability Code, complete outputs, the frozen plan and source hashes are in the public revision repository. Original credit evidence is pinned to `03166e9`; plan and implementation commits are given above. Public source datasets are cited. [Author to confirm final data-deposit statement and any archival release.]

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process The author used ChatGPT (OpenAI) for study design, code, drafting, the core-revision experiments and verification, and Claude (Anthropic) for earlier experimental extensions, analysis and drafting. Figures were generated from numerical outputs using analysis code. [The author must review, edit and adopt the work and takes full responsibility for the published content.]

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